

Top 10 Open Source Software for Engineering Researchers

This article presents a selection of open source software which the author, an engineering researcher, actually uses in his day-to-day work.



Engineering research demands a computation platform that is free from patents and it is not plagued by proprietary rules. It is all about innovation. Engineering complex systems and making significant contributions to any particular field of research is the key motive to any researcher. Budding researchers should build on existing research platforms without ambiguity. In this context, it is necessary to recognise the importance of open source software and GNU general public licence. The software tools I currently use in my research include Ubuntu-14.04-LTS, which is a Debian distribution built on OS Linux-3.19.0-61-generic kernel. The processor used is Intel N3540, which is a low wattage (TDP) processor with a decent clock frequency. The processor gives considerable battery backup and performance.

The other handy open source software tools are.

Linux

Linux kernel 0.01 was released by Linus Torvalds in 1991. He wrote it in C programming language with a GCC compiler. It had 10,239 lines of code then. Now, it has more than 2 billion lines of code. An estimated 3 billion Android phones with the Linux kernel were sold in 2014. Almost all electronic devices support this code. Supercomputers and servers are run on Linux. They are so stable that upgrades are possible without even shutting them down. In 2005, Linus Torvalds said, "Linux is evolution, not intelligent design." Hence, this portable

module will keep on upgrading into new variants and a feature-rich kernel, with the GNU general public licence and keeping the source code intact.

Figure 1 (courtesy: Wikipedia) will help us to visualise the program flow in Linux. The top portion of the figure is where the human/user interface happens and the bottom portion shows the machine interaction. It is above this other distros like Ubuntu, Fedora, etc are built. The processing speed of a computer is determined by the clock speed. The instruction to cycle that relay on clock speed are obviously fast and efficient. Remember, supercomputers are powered by Linux.

LaTeX

In research, publication is a must in order to validate the work. With LaTeX, you can write a journal or conference paper without worrying about its alignment. LaTeX is capable of aligning books, your thesis or even your presentation slides. Learning LaTeX can be a challenge, but a graphical user interface (GUI) like *Tex Live* can be used for easy learning. The file system in LaTeX is open source. After installation, you can check your software package version in your computer by typing the following command in command line interface (CLI or terminal). The version I use is also given below.

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hithu@linux: $ latex -- version
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